



HABIB UNIVERSITY

Data Structures & Algorithms

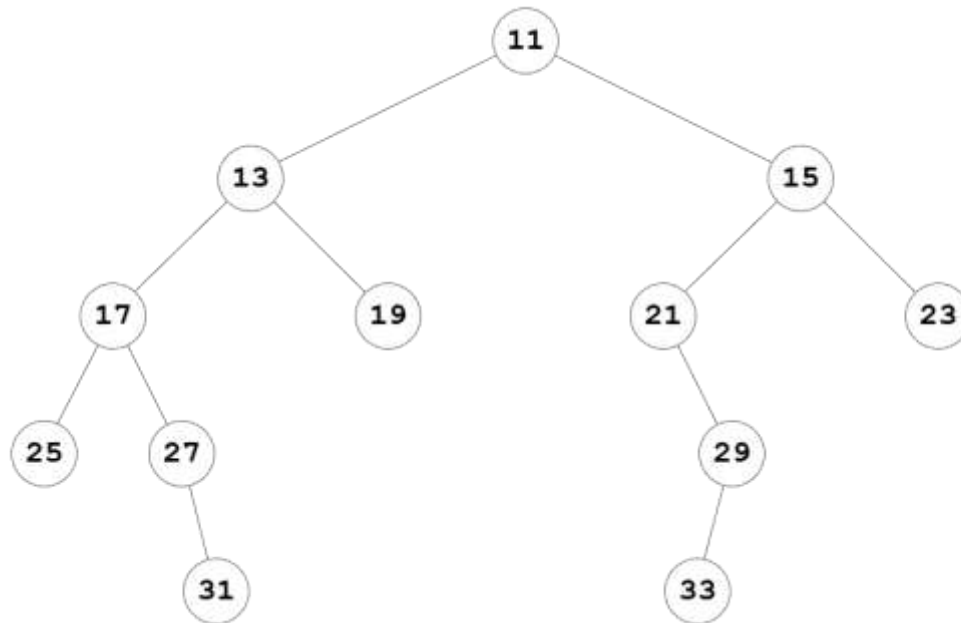
CS/CE 102/171 Spring 2025

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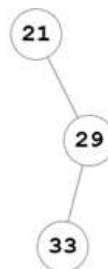
Quiz # 4 Solution

QUESTION: Tree Terminology and Functions

For the given tree, T, answer the following questions:

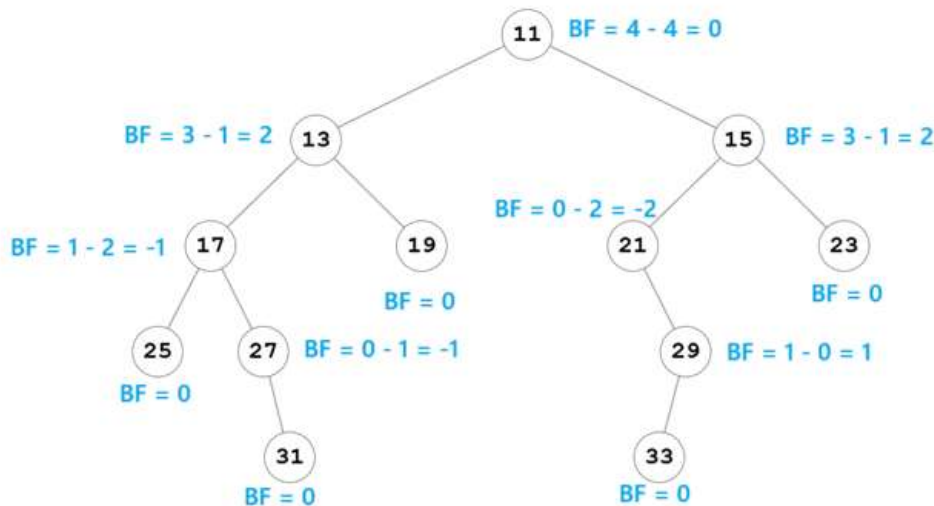


1. degree (33): **1**
2. level (29): **3**
3. height (T, 15): **4**
4. Path from 11 to 31: **11 → 13 → 17 → 27 → 31**
5. List down all the external nodes: **[25, 31, 19, 33, 23]**
6. Draw the left subtree of tree rooted at 15:



7. Siblings of 21: **23**
8. Cousins of 27: **None**

9. Calculate Balance Factor of the given tree, by showing each calculation step to get complete grade. You may solve it directly on the tree given in the question.



10. Is this a balanced binary tree? Give reason for your answer: ***No, because multiple nodes have a balance factor is out of range of -1, 0 and 1***
11. Is it Left-Heavy or Right-Heavy or a Perfectly Balanced Binary Tree? Briefly explain your choice. ***Neither, because it was never balanced to begin with***
12. Inorder Tree Traversal: ***{ 25, 17, 27, 31, 13, 19, 11, 21, 33, 29, 15, 23 }***
13. Preorder Tree Traversal: ***{ 11, 13, 17, 25, 27, 31, 19, 15, 21, 29, 33, 23 }***
14. Array Based Representation of the given tree: ***[11, 13, 15, 17, 19, 21, 23, 25, 27, None, None, None, 29, None, None, None, None, None, 31, None, None, None, None, None, None, 33, None, None, None, None]***
15. Using the Array Representation for Trees, calculate the following: (Show complete working using formulas to prove your answer. Just giving the final answer will result in 0 grade)

- Is Node 27 a leaf node?

For a leaf node, there is no left or right child, so we can check that to deduce if it is a leaf node or not

Position of left child of Node 27 = $2 f(27) + 1 = 2(8) + 1 = 17$

Array[17] = None, which means there is no left child. Check for right child next

Position of right child of Node 27 = $2 f(27) + 2 = 2(8) + 2 = 18$

Array[17] = 31, which means Node 27 has at least one child, hence it cannot be a leaf node

***** For the following 3 questions (Q16, Q17 & Q18), do NOT just write down a number as your answer, but also specify the formula that you used to get the answer. No marks will be awarded for giving ONLY the answer *****

16. For a binary tree with height = 10, total number of nodes possible in the tree is:

$$2^h - 1 = 2^{10} - 1 = 1024 - 1 = 1023$$

17. What are the least number of levels a binary tree with 20 leaves can have?

$$\lceil \log_2 n_L \rceil + 1 = \lceil \log_2 20 \rceil + 1 = \lceil 4.322 \rceil + 1 = 5 + 1 = 6$$

18. For a Binary Tree containing 30 internal nodes, the total number of leaf nodes in the entire tree is:

$$n_L = n_i + 1 = 30 + 1 = 31$$